Gary Feng

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Machine learning and software engineer at Microsoft building production AI and cloud services, with research in computer vision and robotics.

Experience

Machine Learning Engineer, Microsoft | Sep 2024 - Present
Seattle, WA
- Shipped a cross-encoder reranking service that increased click-through rate (CTR) by **12%** in A/B testing, serving **500 requests per second at 90 ms**.
- Built retrieval-augmented generation (RAG) systems with BM25, vector search, and LlamaIndex; evaluated retrieval using GPT-4.5-distilled metrics.
- Deployed a large language model (LLM) evaluation harness using Prometheus-2 (7B) and Azure Machine Learning to score Copilot suggestions, improving offline/online alignment by **8%**.
- Launched a news-intelligence agent for source ingestion, event clustering, deduplication, and ranked summaries; built Azure ML data pipelines for model training and deployment.

Software Engineer, Microsoft | Jul 2022 - Sep 2024
Seattle, WA
- Built and maintained a generally available Kubernetes forecasting service serving **15,000+ clients**, including Dynamics 365 CRM and Microsoft 365 Support Center.
- Built a multi-region recommendation platform on Azure and Docker supporting **20,000+ daily clicks**.
- Led cross-functional model improvements using A/B test results and provided production on-call support, incident diagnosis, and service fixes.

Software Engineer, Skeena Bioenergy Ltd. | May 2020 - Aug 2020
Vancouver, BC, Canada
- Automated PDF-to-SQL data migration with computer vision and built Power BI dashboards from production data.

Technical Skills

Languages: Python, C++, C, C#, SQL

Machine Learning and AI: PyTorch, computer vision, self-supervised learning, vision transformers, LLM evaluation, RAG, information retrieval, search ranking, AI agents, LlamaIndex, Hugging Face

Cloud and Systems: Azure, Azure Machine Learning, Kubernetes, Docker, PostgreSQL, data pipelines, model deployment, A/B testing; **Robotics:** ROS 2, MuJoCo, Gazebo, OpenCV

Research and Projects

I-JEPA for Retinal OCT, NeurIPS 2026 Workshop | 2026
- Pretrained ViT-B/16 on **600,000 OCT scans** with I-JEPA for glaucoma classification; anatomy-shaped masking achieved **0.8947 test AUC** versus 0.8878 for random masking.
- Compared three probe architectures with paired-bootstrap confidence intervals and occlusion attribution; packaged pretrained encoders on Hugging Face. Collaborated with W. T. Lau, Columbia.

CopilotWorldLab, World Models for Robot Manipulation | Jul 2026 - Present
- Reproduced V-JEPA 2-AC for goal-image robot control using model-predictive control in MuJoCo; built a reproducible **500-rollout benchmark** across five tasks and two objects.

MOT17 Multi-Object Tracking | Jan 2026 - Feb 2026
- Combined YOLOv8 and ByteTrack for pedestrian tracking, improving tracking accuracy by **8%** over base YOLOv8.

ROS 2 Robot, Localization and Control | Feb 2025 - May 2025
- Built a differential-drive stack with C++/Python nodes, URDF, Gazebo, and RViz2; fused wheel odometry and simulated IMU data with an extended Kalman filter (EKF).

Publications

Where to Predict in Retinal OCT? Anatomy-Guided Target Selection for I-JEPA.
Y. Feng, W. T. Lau. NeurIPS 2026 Workshop on GenAI for Health (GenAI4Health). OpenReview

Evolution of Humanoid Locomotion Control.
Y. Gu, G. Shi, et al., including **Y. Feng**. Science Robotics, 11(117), 2026. doi:10.1126/scirobotics.aed3973

Education

University of Pennsylvania | Incoming Master of Science in Engineering (M.S.E.), Robotics | Starts 2027

Georgia Institute of Technology | Bachelor of Science (B.S.), Computer Engineering | May 2022